

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Sixth Semester B.Tech (AI&ML) (CBCS) Degree Examinations**

May 2025

AM3236-1 – GENERATIVE ADVERSARIAL NETWORKS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is a common challenge when training GANs?**
A) Overfitting
B) Mode collapse
C) Underfitting
D) Dimensionality reduction
2. **Which of the following is NOT a type of generative model?**
A) Variational Autoencoders (VAEs)
B) Generative Adversarial Networks (GANs)
C) Support Vector Machines (SVMs)
D) Normalizing Flows
3. **How does the generator improve its output?**
A) By minimizing the loss of the discriminator
B) By maximizing the probability of fooling the discriminator
C) By increasing its own loss function
D) By reducing the real data distribution
4. **What is a key feature of DCGANs?**
A) They use fully connected layers
B) They use convolutional layers in both generator and discriminator
C) They do not use activation functions
D) They use autoencoders
5. **What is mode collapse in GANs?**
A) When the generator produces only a few types of outputs
B) When the discriminator overpowers the generator
C) When training time is too long
D) When the generator stops learning completely
6. **Which optimizer is commonly used in GAN training?**
A) SGD
B) Adam
C) RMSprop
D) Newton's Method
7. **In GAN training, how do we deal with vanishing gradients?**
A) Use Leaky ReLU instead of standard ReLU
B) Reduce the number of layers in the model
C) Use a higher learning rate
D) Train only the discriminator
8. **What is the input to the discriminator in a 1D GAN?**
A) A random noise vector
B) Both real and generated 1D functions
C) Only real data
D) A pre-trained model's output
9. **Why is Leaky ReLU preferred over ReLU in a DCGAN discriminator?**
A) Prevents vanishing gradients
B) Increases model complexity
C) Reduces computation time
D) Ensures strict binary classification
10. **What is the purpose of the generator in a DCGAN?**
A) Generate realistic data samples
B) Detect anomalies in the dataset
C) Discriminate between real and fake data
D) Minimize the discriminator's loss
11. **What type of layer is commonly used as the first layer in a DCGAN generator?**
A) Fully connected layer
B) Convolutional layer
C) Transposed convolutional layer
D) Recurrent layer

12. **Why is qualitative evaluation used in GAN performance assessment?**
 A) To compute exact numerical scores B) To analyze generated images visually
 C) To speed up training D) To adjust hyperparameters automatically
13. **Why is batch normalization used in the DCGAN generator?**
 A) Normalize pixel values B) Improve stability and training speed
 C) Increase network depth D) It is computationally expensive
14. **What is a key advantage of using Inception Score (IS) for GAN evaluation?**
 A) Measures sample diversity and image quality B) Computes exact ground-truth labels
 C) Increases training efficiency D) Detects all mode collapse issues
15. **Which regularization technique can help stabilize GAN training?**
 A) Gradient penalty B) Increasing dropout
 C) Removing batch normalization D) Using fixed learning rates
16. **What is the effect of increasing the learning rate of the discriminator too much?**
 A) The generator overfits B) The discriminator dominates, causing training instability
 C) The generator learns faster D) Training time decreases significantly
17. **Why can the standard GAN loss be ineffective?**
 A) The gradients can vanish when the discriminator is too strong B) It requires extremely large datasets
 C) It always produces blurry images D) It only works for black-and-white images
18. **Why can the standard GAN loss be ineffective?**
 A) It only works for black-and-white images B) It requires extremely large datasets
 C) It always produces blurry images D) The gradients can vanish when the discriminator is too strong.
19. **What happens when the generator loss is too high?**
 A) Normalize the data B) The discriminator is too weak
 C) Mode collapse occurs immediately D) The images become perfect
20. **Why are GANs considered brittle?**
 A) They are highly sensitive to hyperparameter tuning B) They do not require large datasets
 C) They never converge D) They only work on grayscale images

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With an example explain the techniques used for Generator and Discriminator model.	8	L*2	1	1
	b) Write a program to up sample the image using transpose convolution operation and elaborate with an example.	8	L2	1	2
2.	a) Write GAN training algorithm and explain it.	8	L2	1	1
	b) Write a program using MNIST dataset for training GAN by mentioning 10 epochs using DCGAN.	8	L3	2	2
3.	a) Provide an overview of what 1D GANs are and how they differ from traditional GANs with the architecture of a typical 1D GAN, including the generator and discriminator components.	8	L2	2	1
	b) Write a program to construct 1D GAN using the function $F(x)=x*x$.	8	L3	2	2
Unit – II					
4.	a) Explain DCGAN training process by considering example.	8	L2	3	1
	b) Discuss the concept of the Mode of collapse in GAN, and suggest strategies to mitigate them.	8	L2	4	1
5.	a) Describe the process of data preprocessing and input preparation for a DCGAN.	8	L2	3	1
	b) Assess the performance of a DCGAN model using the Inception Score.	8	L2	3	1

6. a) Illustrate how qualitative and quantitative evaluation can be done for GAN.
- b) What are the key concerns when analyzing the effectiveness of a generator model?

8 L2 4

8 L2 4

Unit – III

7. a) Describe the limitations of using only the adversarial loss in GANs. How does the vanishing gradient problem relate to this loss, and what solutions have been proposed to address it.
- b) Provide an explanation of regularization methods in GANs, illustrated with an example.
8. a) What is the purpose of the generator loss in a GAN? Explain the differences between common generator loss functions like mean squared error (MSE) and binary cross-entropy (BCE) loss. When might one be preferred over the other.
- b) Discuss the effect of scaling up GANs on their performance, particularly with regard to model size and complexity.

8 L2 5

8 L2 5

8 L3 5

8 L2 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
